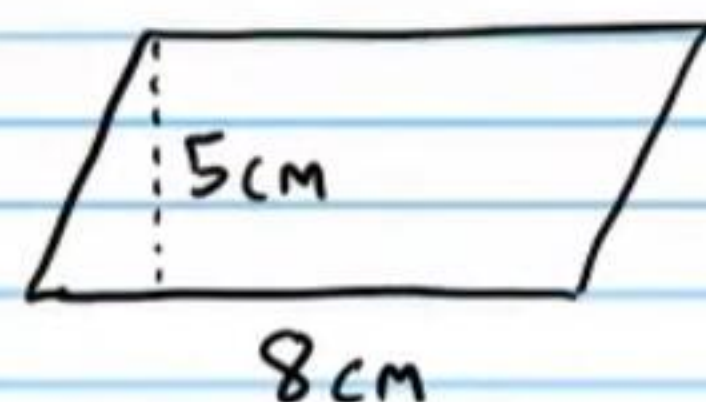


Fundamental Geometry (Part III) Homework

Name
Date
Course

- ① Find the area of the parallelogram below.



$$40 \text{ cm}^2$$

- ② Find the radius of the circle below.

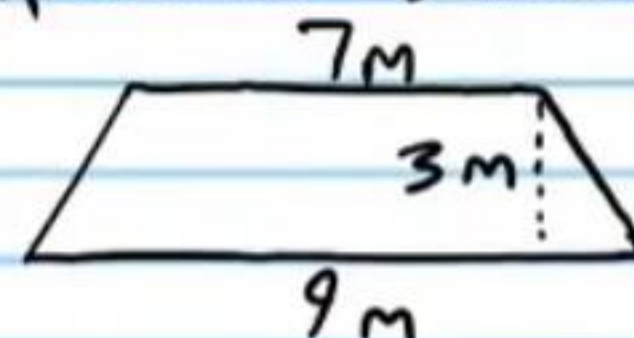


$$6 \text{ ft}$$

- ③ Find the circumference of the circle in problem #2.

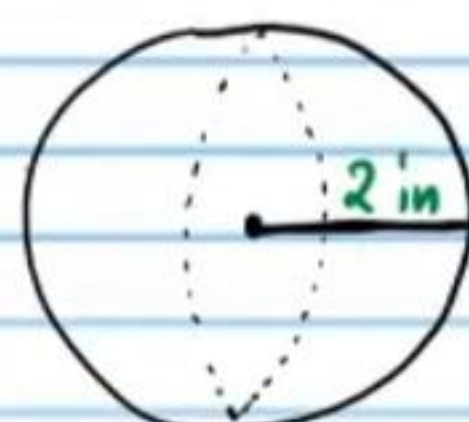
$$12\pi \text{ ft}$$

- ④ Find the area of the trapezoid below.



$$24 \text{ m}^2$$

- ⑤ Find the surface area and volume of the sphere below.



$$A = 16\pi \text{ in}^2$$

$$V = \frac{32\pi}{3} \text{ in}^3$$

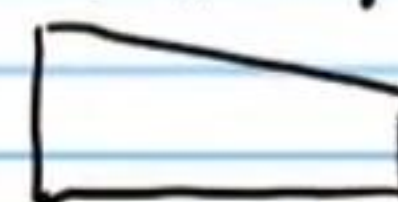
- ⑥ How many sides does a 15-gon have?

15

- ⑦ What is the name of a nine sided polygon? How many angles does it have?

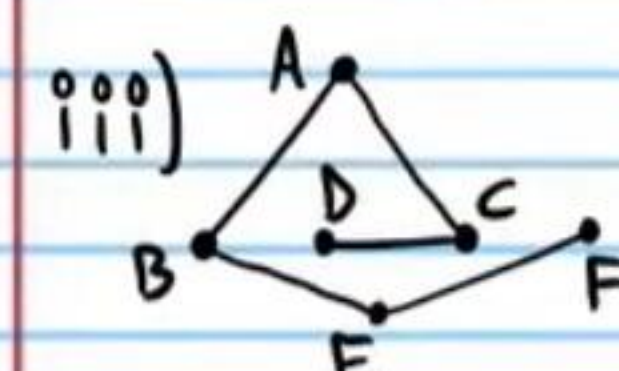
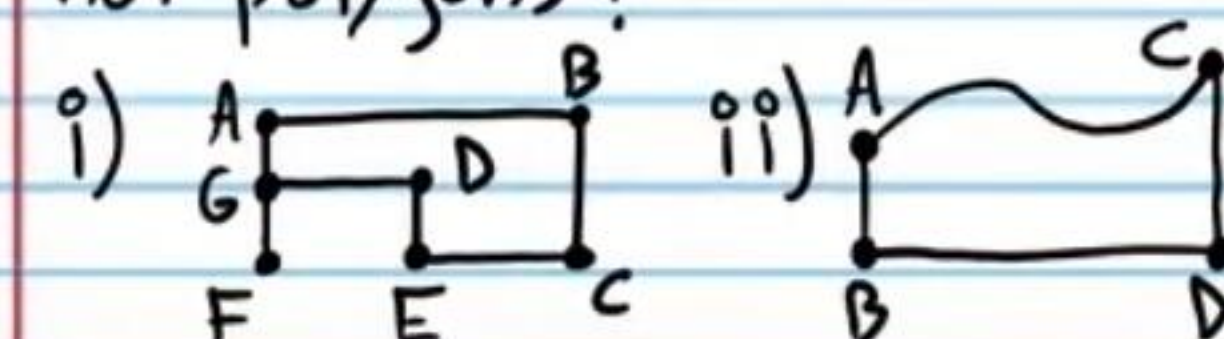
Nonagon
9 angles

- ⑧ What is a name for the polygon below?



Quadrilateral

- ⑨ Why are the figures below not polygons?

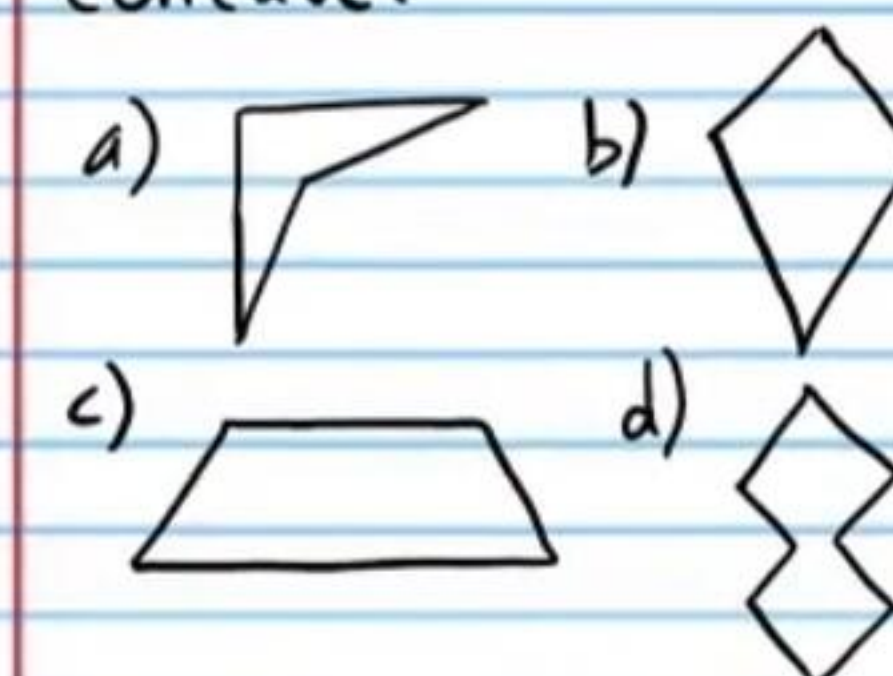


i) Side GD intersects side AF near its center, rather than at endpoint F.

ii) Side AC is curved.

iii) Not closed.

- ⑩ Label the following polygons as convex or concave.



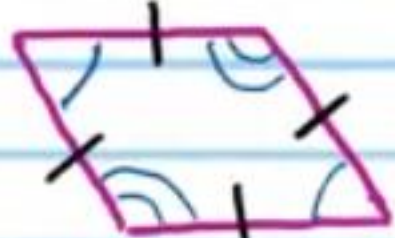
a) Concave
b) Convex
c) Convex
d) Concave

⑪ a) What is a regular polygon?

b) Is an equilateral polygon also a regular polygon?

a) A polygon in which all angles and sides are congruent.

b) Not always. An equilateral polygon may have angles that are not all congruent. Example below.



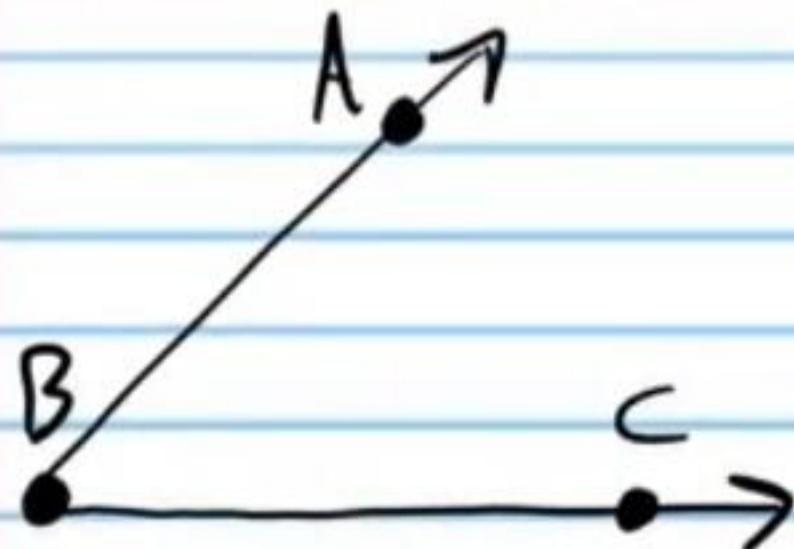
Use a compass and ruler for problem #'s 12-15.

⑫ Draw a segment congruent to $\overline{AB}$ below.



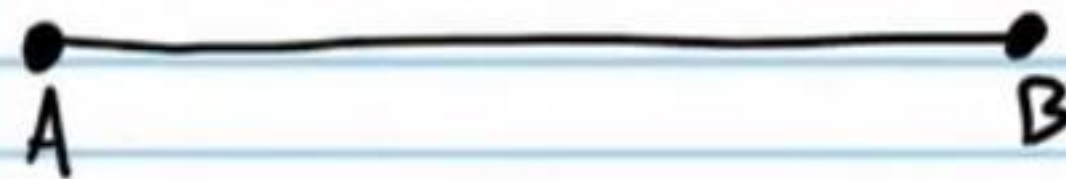
Should look exactly the same as $\overline{AB}$.

⑬ Draw an angle congruent to $\angle ABC$ below.

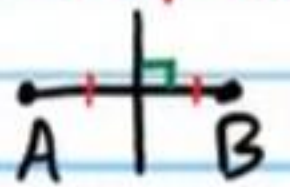


Should look exactly the same as $\angle ABC$.

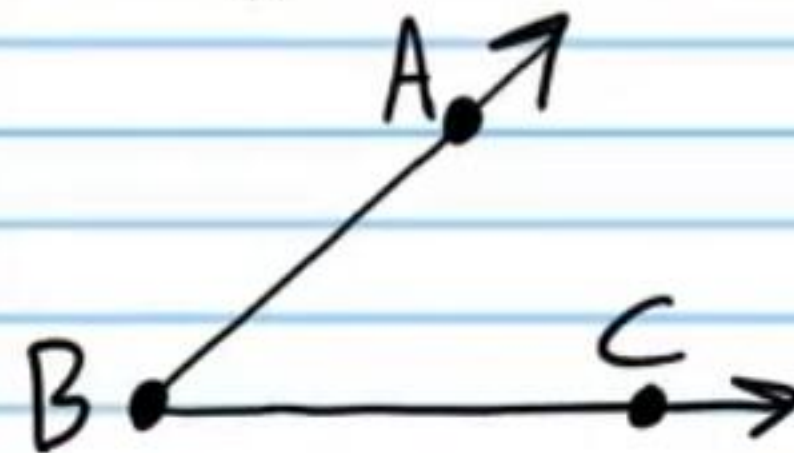
⑭ Draw a perpendicular segment bisector through $\overline{AB}$ below.



Should be a 90° angle and crossing at the **very center**.



⑮ Draw an angle bisector through $\angle ABC$ below.



Should appear to cut $\angle ABC$ in half.

